Material cases: Worked solutions

> Reference: Jennifer Rupp, Daniele Ielmini, Ilia Valov (eds.), Resistive Switching: Oxide Materials, Mechanisms, Devices and Operations, Springer, 2022. DOI: 10.1007/978-3-030-42424-4.

E01 Diffusion scale

Convert $20\,\mathrm{nm} = 2 \times 10^{-6}\,\mathrm{cm}$. From $\langle x^2 \rangle = 2Dt$, $t = (2 \times 10^{-6})^2/(2 \times 10^{-21}) = 2 \times 10^{9}\,\mathrm{s} \approx 63.4$ years. This is the chosen free-diffusion scale, not a SET measurement: electric-field drift, interfaces and film transport are excluded.

E02 Parallel RC

$RC = (10^{6})(10 \times 10^{-12}) = 10^{-5}\,\mathrm{s} = 10\,\mu\mathrm{s}$. Since $Z = R/(1 + j\omega RC)$, the maximum of $-\operatorname{Im} Z$ occurs at $\omega RC = 1$, so $f = 1/(2\pi RC) = 15.9\,\mathrm{kHz}$. No: distinct physical structures may have a similar equivalent impedance.

E03 Weibull probability and median

$F(0.8) = 1 - e^{-(0.8)^4} = 1 - e^{-0.4096} = 0.3361$. With $F(x_{0.5}) = 0.5$, $x_{0.5} = 1[-\ln(0.5)]^{1/4} = 0.9124\,\mathrm{V}$. The result is conditional on the assumed model and its parameters.

E04 Weibull quantiles and data

$x_p = \eta[-\ln(1-p)]^{1/4}$ gives $x_{0.1} = 0.5697\,\mathrm{V}$ and $x_{0.9} = 1.2318\,\mathrm{V}$. Successive cycles share device geometry and history, so they can be correlated; report device count separately, together with censoring, selection, binning and uncertainty as appropriate.

E05 Heating and scaling

$P = V^2/R = 0.81/150 = 0.0054\,\mathrm{W} = 5.4\,\mathrm{mW}$ and $\Delta T = R_{\mathrm{th}} P = 162\,\mathrm{K}$. If RESET is modeled by constant ΔT_c, then $V_{\mathrm{RESET}}^2 = \Delta T_c R/R_{\mathrm{th}}$. Independence of R additionally requires the ratio R/R_{th} to stay fixed; steady operation and the lumped model must also be appropriate.

E06 HfO_2/Ti aging evidence

The ratio is smaller by $10^4/10^2 = 100$. One ratio per time does not separately identify either absolute resistance or an oxygen trajectory. Depth-profile XPS may support a change in elemental or chemical distribution, subject to depth resolution, sputter effects and sample differences; it is not direct real-time tracking of an operating device.

E07 Classical nucleation

Differentiation gives $r^* = 2\gamma/|\Delta g_v| = 0.06/(3 \times 10^7) = 2.0 \times 10^{-9}\,\mathrm{m} = 2.0\,\mathrm{nm}$. Substitution yields $\Delta G^* = f16\pi\gamma^3/(3|\Delta g_v|^2) = 5.03 \times 10^{-20}\,\mathrm{J} \approx 0.314\,\mathrm{eV}$. The geometry and constant material parameters are model assumptions.

E08 Thermal factor and rate

$\Delta G^*/k_B T \approx 0.314/0.02585 = 12.15$, so the factor is approximately 5.3×10^{-6}. It is dimensionless. A rate also requires a kinetic prefactor and a number of available sites; atomistic cluster size, interface geometry and nonequilibrium transport can invalidate the classical approximation.

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E09 Fast pulse

$I_C = C\,dV/dt = 0.2 \times 10^{-12} \times 10^{9} = 2 \times 10^{-4}\,\mathrm{A} = 0.2\,\mathrm{mA}$. With the specified definitions the SET time is $1.2\,\mathrm{ns}$, not the time of the earlier capacitive current. Energy depends on the full time-dependent voltage and device current, $E = \int V(t) I_{\mathrm{device}}(t) dt$, and on the chosen interval and parasitic-energy treatment; a peak does not supply that integral.

E10 Three-terminal switch

A two-terminal device bridges its two electrodes. In the three-terminal example, the gate supplies or regulates ions; a separately biased source–drain pair probes and can be bridged by a conductive path. A path to the gate is a different, unwanted short. Ion migration and redox need not follow a MOSFET's field-effect channel mechanism, even though both devices have three terminals.

E11 CBRAM reactions and humidity

Illustrative half-reactions are $\mathrm{Cu} \rightarrow \mathrm{Cu}^{+} + e^{-}$ and $\mathrm{Cu}^{+} + e^{-} \rightarrow \mathrm{Cu}$. Growth depends on transport, nucleation and local electrochemical reactions, so polarity alone is insufficient. A changing ion concentration and approximately unchanged estimated diffusivity do not demonstrate increased mobility; the reported behavior is specific to the sample and conditions and cannot establish a universal humidity trend.

E12 Evidence and alternatives

Hysteresis at the stated temperatures is the observation, not identification of a mechanism. Under matched pulse conditions, vary temperature or the thermal boundary and compare timing and energy; a systematic trend can support a thermally assisted process but is not unique evidence. Compare synchronized electrical cycling with composition-sensitive imaging and matched controls: correlation can support metal rearrangement but probe perturbation and spatial resolution remain limiting. Measure a dummy structure or vary edge rate and bandwidth to estimate capacitance-driven transients; this can reveal parasitic contributions but does not on its own identify the switching material. Multiple independent controls are needed to distinguish competing models.

E13 Conditional field drift

$\mu = D/(k_B T/e) = 10^{-21}/0.02585 = 3.868 \times 10^{-20}\,\mathrm{cm^2/(V\,s)}$. Thus $v_d = \mu E = 3.868 \times 10^{-15}\,\mathrm{cm/s}$ and, with $20\,\mathrm{nm} = 2 \times 10^{-6}\,\mathrm{cm}$, $L/v_d = 5.17 \times 10^{8}\,\mathrm{s}$. This deliberately idealized scale omits diffusion, contact reactions and a varying field; linear mobility and Einstein's relation are assumed, and the field value alone does not establish them. In a hopping model, $qaE \ll k_B T$ is one condition to check. This is not a measured SET time.

E14 RESET current bins

Using $I = 0.9/R$, the currents are respectively 9, 6, 4.5, 3.6, 3, and $2.25\,\mathrm{mA}$. In the steady model $\Delta T = R_{\mathrm{th}} V^2/R$, the same critical temperature rise produces $|V| = \sqrt{\Delta T_c R/R_{\mathrm{th}}}$. A constant peak voltage therefore requires a common ΔT_c and approximately constant R_{th}/R, as well as an adequate lumped, steady model. The six values are assumptions, not experimental data.

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E15 Ti-thickness comparison

$(5 \times 10^4)/10 = 5 \times 10^3$: the first ratio is 5,000 times the last. These are group-level, condition-specific ratios; they do not independently determine HRS and LRS or isolate Ti thickness from fabrication, sampling and storage. Different device batches, read polarities and added interlayers cannot be silently treated as the same experiment. XPS profiling adds support for redistribution but can be affected by preparation and sputtering.

E16 Competing timing laws

At 1, 1.5 and 2 V, $\ln(t/\mathrm{ns}) = 4, 3, 2$, giving $e^4 = 54.6$, $e^3 = 20.1$, and $e^2 = 7.39\,\mathrm{ns}$. The condition $\Delta G_0 > 0.10\,\mathrm{eV}$ keeps the assumed nucleation barrier positive over this range. For the assumed migration law, $\ln t = \ln(L/v_0) - [a/(2L(k_BT/e))]V$, and the magnitude of the slope is $(0.3/3)/(2 \times 0.025) = 2\,\mathrm{V}^{-1}$. Equal slopes do not select a unique mechanism; vary temperature, gap thickness and other controlled conditions and test independent composition or timing evidence.

E17 Hypothetical multilevel programming

The model gives $0.5/(5\,\mu\mathrm{A}) = 100\,\mathrm{k\Omega}$, then $50\,\mathrm{k\Omega}$ and $25\,\mathrm{k\Omega}$. Ohm's law at 0.1 V yields respectively 1, 2 and 4 µA. These are nominal assumed values, not experimental data. Overlap of programming and read distributions, drift during the required retention time, noise and read disturbance can make thresholds unreliable even when nominal values differ.

E18 Radiation and evidence

Similar distributions and programming curves support little detectable change for these sampled structures, doses, read and cycling conditions. Ten cycles on one cell are not ten independent devices; the small numbers and plotted overlap cannot exclude rare failure or a small shift without uncertainty analysis. Total ionizing dose does not measure every radiation mechanism; other dose rates, materials, particles, peripheral circuitry, endurance and long-term retention need separate tests. $\mathrm{Mrad(SiO_2)}$ specifies the reference material for the stated dose.

E19 Ta_2O_5 film density and humidity

For each film use the same temperature and electrical protocols, cycle dry → humid → dry, include unprogrammed and repeated-cell controls, and record transient conductance as well as post-bias retention. Recovery supports a reversible environmental contribution, but density and deposition can alter multiple properties. The proposed proton path remains a hypothesis; isotope-sensitive or other independent carrier tests and controls are needed. The reported 50% boundary belongs to this tested structure and procedure, not all devices. The 5–85% RH, 100 mV/s cyclic-voltammetry comparison is a separate study; its fitted ion concentration and diffusion coefficient depend on its own sample, protocol and model, and cannot be assigned to either density-comparison film without evidence.

Material cases: Worked solutions (continued)

E20 Spectroscopic and impedance evidence

Spectral changes consistent with particles and changing locations support a local structural interpretation, but the plasmonic response depends on the electrodes and optical geometry and is not a direct picture. For the programmed Cu/SiO_2 cell, nearly unchanged fitted C despite large R changes supports conduction that occupies a small fraction of the capacitor area, within the chosen RC model. A different microscopic structure or more than one circuit can yield similar response. Repeat the fit across frequency and area with controls and combine synchronized, composition-sensitive imaging or other independent measurements; account for probe and sample perturbation.

E21 Synaptic pulse readout

By the series assumption, $I = V_2/R_s$ and $V_{\mathrm{dev}} = V_1 - V_2$. Thus $R_{\mathrm{dev}} = V_{\mathrm{dev}}/I = R_s(V_1/V_2 - 1)$. With $V_1 = 1\,\mathrm{V}$ and $R_s = 10\,\mathrm{k\Omega}$, the three values are $10(5-1) = 40$, $10(4-1) = 30$, and $10(3.333\ldots - 1) = 23.33\,\mathrm{k\Omega}$. Their reciprocals are 25, 33.33 and 42.86 µS. This is a hypothetical three-sample calculation: it is not the reported experiment's ten-pulse group, pulse amplitudes or measured values. Programming can change the device while it is sampled, so synchronize actual voltages and account for bandwidth and capacitive current; test retention and read disturbance under a separately specified read protocol.

E22 Pulse energy and endurance

$E = VIt = (0.8)(50 \times 10^{-6})(10 \times 10^{-9}) = 4 \times 10^{-13}\,\mathrm{J} = 0.4\,\mathrm{pJ}$ per pulse. Two pulses per cycle for 10^4 cycles give $2 \times 10^4 \times 0.4\,\mathrm{pJ} = 8\,\mathrm{nJ}$. These are hypothetical constant waveforms and omit peripheral, read and parasitic energy. Measure voltage and current waveforms including transients, device temperature where feasible, HRS/LRS distributions over cycles, failure thresholds and sample-to-sample variability. Energy alone does not identify damage or yield a failure distribution; the source's reported 10^4 cycles without obvious degradation applies to its particular tested 1T1R device and protocol.